

Princeton University
Mathematical Intermediate Microeconomics (ECO 310)
Midterm Spring 2022 Solutions

Instructions: You have 80 minutes to complete this exam, plus 10 minutes to upload your answers. Each part of each question (Q1, Q2, and Q3) is worth 10, for a total of 100. **Show your work and explain your answers for full credit!**

- The exam is open-book, and calculators are permitted. However, please work **alone**.
- **If you are unsure whether you're interpreting a question correctly, state your interpretation and proceed (or email me if you prefer).**

The honor code applies, as usual.

Q1: Proofs and General Concepts

- a. Sally has preferences over $X = \mathbb{R}^2$ represented by $u(x_1, x_2) = \max\{x_1, x_2\}$. Are her preferences (i) rational? (ii) convex?

Answer: (i) they are rational, since the existence of a utility representation ensures that they are complete and transitive. (This is enough of an answer, but more precisely: for any two bundles x and y , either $u(x) \geq u(y)$, in which case $x \succsim y$, or $u(y) > u(x)$, in which case $y \succ x$; so we have completeness. And if $x \succsim y$ and $y \succsim z$, thus $u(x) \geq u(y)$ and $u(y) \geq u(z)$, then $x \succsim z$; so transitivity holds too). (ii) They are not convex. For example, consider the bundles $(1, 3)$ and $(3, 1)$; convexity requires that any weighted average, in particular the 50-50 weighted average, be weakly preferred to the worst of these bundles. But $u(1, 3) = u(3, 1) = \max\{1, 3\} = 3$, whereas $u(\frac{1}{2}(1, 3) + \frac{1}{2}(3, 1)) = u(2, 2) = 2$.

- b. Now Sally tells you that she has a utility function $u : X \rightarrow \mathbb{R}$, and, from any menu $A \subseteq X$, chooses the option in A with the lowest utility ($C(A) = \arg \min_{x \in A} u(x)$). Do her choices satisfy IIA? (proof or counterexample)

Answer: yes. Argument #1: let $a = C(A)$. Thus $a = \arg \min_{x \in A} u(x)$. But then for any subset $B \subseteq A$ containing a , since we already know that $u(a) \leq u(x) \forall x \in A$, in particular it holds that $u(a) \leq u(x) \forall x \in B \subseteq A$; thus $a \in C(B)$, verifying IIA.

Argument #2: let v be the negative of Sally's utility function. Then Sally's choices are the ones that maximize utility function v , and we know that such rationalizable choices satisfy IIA.

- c. Using only the assumption that utility of expected wealth beats expected utility, prove that a gamble with 50-50 odds of winning or losing \$100 is preferred to a gamble with 50-50 odds of winning or losing \$200.

Answer: we want to show that

$$\frac{1}{2}u(w_0 - 100) + \frac{1}{2}u(w_0 + 100) > \frac{1}{2}u(w_0 - 200) + \frac{1}{2}u(w_0 + 200) \quad (*)$$

Since utility of expected wealth beats EU, we know that

$$\begin{aligned} u(w_0 - 100) &> \frac{1}{2}u(w_0 - 200) + \frac{1}{2}u(w_0) \\ u(w_0 + 100) &> \frac{1}{2}u(w_0) + \frac{1}{2}u(w_0 + 200) \end{aligned}$$

Substituting into the LHS of (*), it suffices to prove that

$$\begin{aligned} &\frac{1}{4}u(w_0 - 200) + \frac{1}{4}u(w_0 + 200) + \frac{1}{2}u(w_0) > \frac{1}{2}u(w_0 - 200) + \frac{1}{2}u(w_0 + 200) \\ \Leftrightarrow &u(w_0) > \frac{1}{2}u(w_0 - 200) + \frac{1}{2}u(w_0 + 200) \end{aligned}$$

which holds again using the fact that utility of expected wealth beats EU.

Q2: Demand

Homer Simpson's preferences over beer (B) and leisure (L) are given by

$$u(B, L) = \min\{B, L\}$$

He spends 8h per day passed out, but can work as many of the remaining 16 hours as he wishes, at a flat wage w /hour. Marge also gives him an allowance of N per day. He buys cheap beer, with price $p_B = 1$.

- a. Find his optimal labor supply, $\ell(w)$. (i.e., how many hours will he optimally work at wage w ?)

Answer: he wants to maximize $\min\{B, L\}$ subject to budget constraint $B = w(16 - L) + N$. The optimal spending condition is $B = L$ for these perfect complement preferences; substituting into the budget constraint yields

$$\begin{aligned} L &= w(16 - L) + N \\ \Rightarrow L &= \frac{16w + N}{1 + w} \end{aligned}$$

Finally, since Homer has 16 hours per day to allocate between labor and leisure, this gives optimal labor:

$$\ell(w) = 16 - L = 16 - \frac{16w + N}{1 + w} = \frac{16 - N}{1 + w}$$

(This holds provided that $N \leq 16$; if $N > 16$, then Homer chooses a corner solution with no labor, 16 hours of leisure, and then he spends his allowance on beer, so $B = N$).

- b. Calculate his compensating variation if the price of beer doubles.

Answer: with perfect complement preferences, there is no reoptimization for a compensated price change: the optimal spending condition is still $B = L$, and the most efficient way to maintain original utility is to buy exactly the original bundle. The compensation required for this, i.e. the compensating variation, is then the increased cost of his original quantity of beers, which is $2B^0 - 1B^0 = B^0$. Most-to-all credit for this, but finally, the actual original # beers can be obtained from (a): from the budget constraint,

$$B^0 = w(16 - L) + N = \begin{cases} w \frac{16 - N}{1 + w} + N & \text{if } N \leq 16 \\ N & \text{if } N > 16 \end{cases} = \begin{cases} \frac{16w + N}{1 + w} & \text{if } N \leq 16 \\ N & \text{if } N > 16 \end{cases}$$

- c. How much allowance (N) does Homer need at beer price p_B and wage rate w to get utility u^* ? (Assume $u^* \leq 16$)

Answer: with perfect complements, optimality requires $B = L$, and so then to get utility u^* , Homer needs $B = L = u^*$. Substituting into his budget constraint $p_B B = w(16 - L) + N$, the allowance N required to afford this satisfies

$$N = p_B B - w(16 - L) = p_B u^* - w(16 - u^*) = (p_B + w)u^* - 16w$$

Q3: Expected Utility

Sam has 60 minutes to complete his midterm, which has both an easy question and a hard question. For parts (a)-(c), assume that he can choose how to split his time between the two questions. Every minute spent on the easy question earns a guaranteed point. Points on the hard question are uncertain, as Sam is not sure whether he'll figure it out: he expects that if he spends x minutes on it, he'll get $2x$ points with probability p , and zero points otherwise.

- a. Write out Sam's expected utility if he spends x minutes on the hard question and the remaining $60 - x$ minutes on the easy question, assuming utility $\sqrt{T}$ from point total T .

Answer: with probability p , Sam gets the hard question right, for point total $60 - x + 2x = 60 + x$. With probability $1 - p$, he gets it wrong, for point total $60 - x$ (easy question only). Thus

$$EU = p\sqrt{60 + x} + (1 - p)\sqrt{60 - x}$$

- b. Suppose (for this part only) that $p < \frac{1}{2}$. Show that in expectation, each minute spent on the hard question earns less than a point, and explain why this implies that Sam should devote the entire 60 minutes to the easy question.

Answer: x minutes on the hard question yield expected point total $p(2x) + (1 - p)(0) = 2px$; if $p < 1$, then this is smaller than x (so each minute yields less than a point in expectation). But each minute on the easy question yields a guaranteed point. Also note that Sam is risk-averse, since $\sqrt{T}$ is a concave function. So since the easy question yields a higher expected # points, AND is better in terms of having no risk, Sam strictly prefers to allocate all points to the easy question.

- c. Assuming now that $p > \frac{1}{2}$, find the optimal # minutes x that Sam should spend on the hard question. (Partial minutes **are** allowed).

Answer: Sam chooses x to maximize the EU expression in (a). The FOC is

$$\begin{aligned} \frac{p}{2\sqrt{60 + x}} - \frac{1 - p}{2\sqrt{60 - x}} &= 0 \Rightarrow \frac{p}{1 - p} = \sqrt{\frac{60 + x}{60 - x}} \\ &\Rightarrow (60 - x)p^2 = (60 + x)(1 - p)^2 \\ &\Rightarrow 60(p^2 - (1 - p)^2) = x(p^2 + (1 - p)^2) \\ &\Rightarrow x = 60 \frac{p^2 - (1 - p)^2}{p^2 + (1 - p)^2} \end{aligned}$$

- d. Now modify the question slightly. Suppose Sam has to choose at the outset to do the hard question **OR** the easy question, he cannot attempt both. For what values of p will he choose the hard question?

Answer: from (a), Sam can choose either

$$\begin{aligned} x &= 0 \Rightarrow EU = \sqrt{60}, \text{ or} \\ x &= 60 \Rightarrow EU = p\sqrt{120} \end{aligned}$$

So he prefers all points on the hard question whenever $x = 60$ is better, i.e. $p > \frac{\sqrt{60}}{\sqrt{120}}$.